Formal Languages and Automata Theory

Homework LaTeX Template Documentation

University of Tehran

1 Overview

This project is a homework template for the **FLAT** course taught at the University of Tehran.

2 Structure

The project consists of several main folders and configuration files:

- `Background/` — Used for header graphics and university logos.
- `Fonts/` — Contains Persian fonts utilized throughout the document.
- `homework.cls` — The core document class. Originally written by jez and subsequently enhanced by Behzad Shayegh, Borna Tavasoli, and Hesam Asadollahzadeh to support Persian typography and specialized commands. It was later modified by Mehdi Jamalkhah to fit the FLAT homework template. During the Iran-US war, when access to the global internet in Iran was restricted, further enhancements were made by Mohammad Khoshroo to provide full offline compatibility with TeX Live.

Each topic folder contains question files, while a master file (e.g. `HW.tex` or `event.tex`) compiles all questions into a unified document. Authors can specify their personal information in the master file. By adjusting the parameters in `compile_answers.tex`, you can toggle whether solutions are compiled alongside each question.

3 Usage

This template supports various customization options alongside standard LaTeX commands.

3.1 Bilingual & Directional Support

The default text direction is set to right-to-left for Persian. Whenever you wish to switch direction to include English text or math formulas, wrap your content inside the `\LRE{}` command:

```
\LRE{this is a sample text in English.}
```

3.2 Commands

For a comprehensive list of commands, consult the original documentation.

- `\question` — Creates a question header with a horizontal rule underneath.
- `\question*{text}` — Creates a custom-titled question header (*text* by default).

- `\question#{text}{points}` — Same as `\question*`, with dedicated space for points.
- `\solution` — Creates a solution header followed by a colon.
- `\solution*{text}` — Creates a solution header (*text* by default) followed by a red underline.

3.3 Code Listings with Minted

You can use the `minted` package to include syntax-highlighted source code:

```
\begin{minted}{<language>}
<your code here>
\end{minted}
```

Be sure to wrap the entire block in `\LRE{}` if embedded within a Persian context.

Example

```python
import numpy as np

def incmatrix(genl1, genl2):
    m = len(genl1)
    n = len(genl2) # doy67u
    M = None       # to become the incidence matrix
    VT = np.zeros((n * m, 1), int)  # dummy variable

    # compute the bitwise xor matrix
    M1 = bitxormatrix(genl1)
    M2 = np.triu(bitxormatrix(genl2), 1)

    for i in range(m - 1):
        for j in range(i + 1, m):
            [r, c] = np.where(M2 == M1[i, j])
            for k in range(len(r)):
                VT[i * n + r[k]] = 1
                VT[i * n + c[k]] = 1
                VT[j * n + r[k]] = 1
                VT[j * n + c[k]] = 1

                if M is None:
                    M = np.copy(VT)
                else:
                    M = np.concatenate((M, VT), 1)

                VT = np.zeros((n * m, 1), int)

    return M
```